Almost sure convergence of higher-dimensional LFPP

An upgrade from convergence in probability

Research manuscript

Abstract

We prove an almost-sure upgrade theorem for subcritical exponential metrics of the whole-space log-correlated Gaussian field in dimension $d \geq 3$. The input is a strong metric assignment and locally uniform convergence in probability of median-normalized convolution LFPP to that assignment. The conclusion is locally uniform convergence almost surely along the full continuum of mollification scales. The argument combines shell independence with a comparison of paths in a closed annulus and paths in a larger open annulus. Internal sphere diameters bound the portions of a geodesic not covered by the sharp comparison, giving a fixed contraction of the comparison constant. A deterministic selection of normalization windows then gives summable comparison errors. Microscopic endpoint estimates and confinement of minimizing paths permit interpolation between cutoff scales.

Dependency. The strong characterization and full convergence in probability in the earlier higher-dimensional metric manuscript [5] are inputs here, not conclusions. That manuscript labels its uniqueness proof as proposed. The theorem below proves the almost-sure upgrade conditional on precisely the input stated in Assumption 1.1; it does not independently certify the earlier uniqueness proof.

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1 Introduction

Fix $d \geq 3$ and $\xi > 0$. Let h be the whole-space log-correlated Gaussian field, normalized by $h_1(0) = 0$, where $h_r(z)$ denotes its spherical average. Our covariance convention is

$$\mathrm{Cov}((h,\varphi),(h,\psi)) = \int_{\mathbb{R}^d}\int_{\mathbb{R}^d} \varphi(x)\psi(y)\log\frac{1}{|x-y|}\,dx\,dy\,, \tag{1.1}$$

for smooth compactly supported test functions of integral zero. In dimension greater than two this is not the Gaussian free field.

We use the convolution kernels in [3, Remark 1.1 and Lemma 2.9]. Choose a real, radial function $\kappa \in C_c^\infty(\mathbb{R}^d)$ with $\int \kappa^2 = 1$, and define K by

$$\widehat{K}(\zeta) = \left(\frac{2\pi^{d/2}}{\Gamma(d/2)}\int_{|\zeta|}^\infty t^{d-1}|\widehat{\kappa}(t)|^2\,dt\right)^{1/2}. \tag{1.2}$$

Here $\widehat{f}(\zeta) = \int e^{-2\pi i x\cdot\zeta} f(x)\,dx$, and the radial function $\widehat{\kappa}$ is evaluated at a radius. The kernel K is real and radial, has integral one, and satisfies $|K(x)| \leq C(1+|x|)^{-(2d-1)}$. It need not be nonnegative. Set $K_\epsilon(x) = \epsilon^{-d}K(x/\epsilon)$ and $h_\epsilon^* = h * K_\epsilon$. Define

$$D_h^\epsilon(x,y) = \inf_{P:x\to y}\int_0^1 e^{\xi h_\epsilon^*(P(t))}|P'(t)|\,dt\,, \tag{1.3}$$

where the infimum is over piecewise continuously differentiable paths in $\mathbb{R}^d$. Throughout, the deterministic normalization is the lower median

$$\mathsf{a}_\epsilon = \inf\{a > 0 : \mathbb{P}[D_h^\epsilon(0,e_1) \leq a] \geq 1/2\}\,. \tag{1.4}$$

All approximations are functions of the same pinned field h.

For a metric D, the length of a continuous path is the supremum of the sums of successive distances over finite partitions. We write $D(x,y;U)$ for the infimum of lengths of paths contained in U. We use this notation also for a closed set, with the value $+\infty$ if there is no rectifiable path. Distances between sets are infima over endpoint pairs. Let $B_r(z)$ be the Euclidean open ball, let $S_r(z) = \partial B_r(z)$, and set $A_{a,b}(z) = B_b(z) \setminus \overline{B}_a(z)$. Omitted centers are zero. For a set U, write $B_r(U) = \bigcup_{z\in U} B_r(z)$.

Assumption 1.1 (Strong metric and convergence in probability)**.** There is a measurable metric assignment $g \mapsto D_g$, defined for a whole-space log-correlated Gaussian field plus a continuous function, with the following properties.

(i) Almost surely, D_g is a finite continuous length metric inducing the Euclidean topology.

(ii) For every deterministic open set U, $D_g(\cdot,\cdot;U)$ is measurable with respect to $g|_U$.

(iii) Almost surely, simultaneously for continuous f, $D_{g+f} = e^{\xi f}\cdot D_g$, where the right side minimizes $\int e^{\xi f}\,ds_{D_g}$ over paths.

(iv) For every fixed $r > 0$ and $z \in \mathbb{R}^d$, almost surely for all u, v,

$$D_h(z+ru, z+rv) = r^{\xi Q}e^{\xi h_r(z)}D_{h^{z,r}}(u,v)\,, \qquad h^{z,r} = h(z+r\cdot) - h_r(z)\,, \tag{1.5}$$

where $Q = Q(\xi) > \sqrt{2d}$ is the LFPP distance exponent. The corresponding covariance identities hold for the assignment after continuous perturbations.

(v) On the original field probability space,

$$\mathsf{a}_\epsilon^{-1} D_h^\epsilon \longrightarrow D_h \quad \text{in probability, locally uniformly on } \mathbb{R}^d \times \mathbb{R}^d \,. \tag{1.6}$$

The law of $h^{z,r}$ in (1.5) is the law of h. No simultaneous covariance event over all real r, z is assumed. The proof uses only countably many deterministic centers and radii at each of countably many cutoffs. The four metric axioms and (1.6) are exactly the conclusions needed from the earlier characterization; there is no additional tightness axiom in Assumption 1.1.

Theorem 1.2. *Under Assumption 1.1, almost surely, for every compact $F \subset \mathbb{R}^d$,*

$$\lim_{\epsilon \downarrow 0} \sup_{x,y \in F} \left| \mathsf{a}_\epsilon^{-1} D_h^\epsilon(x,y) - D_h(x,y) \right| = 0 \,. \tag{1.7}$$

The limit is along all positive ϵ tending to zero, for the fixed parameter ξ and fixed kernel K above.

Devlin proved the analogous almost-sure assertion in dimension two [4, Theorem 1.1]. We follow the broad strategy of improving a coarse comparison, selecting favorable normalization scales, and interpolating between a summable sequence of cutoffs. The geometric comparison below is proved directly in $\mathbb{R}^d$: a sphere need not contain a rectifiable separating loop. Instead, its diameter in a larger annulus controls the time a geodesic can spend between successive annular crossings.

There are two reasons that a direct Borel–Cantelli argument from (1.6) does not suffice. First, convergence in probability gives no summable rate. Second, when a unit-scale approximation is transported to a radius r, its deterministic normalization differs from that of D_h by

$$c_\epsilon(r) = \frac{r \mathsf{a}_{\epsilon/r}}{r^{\xi Q} \mathsf{a}_\epsilon} \,. \tag{1.8}$$

Convergence of this ratio for a fixed r does not control radii which themselves tend to zero with ϵ.

Section 2 records the field estimates and coarse comparison used in the proof. Section 3 obtains a local comparison inside a buffered annulus and amplifies its probability using shell independence. The geodesic decomposition in Section 4 improves a comparison constant by a fixed fraction. Section 5 applies this contraction to deterministic windows of radii and obtains a sharp comparison with polynomially small failure probability. Finally, Section 6 removes the endpoint balls and proves convergence along the full continuum of cutoffs.

2 Preliminaries

We first separate local metric convergence from control of paths escaping to infinity. The latter is needed when replacing the convolution by a local truncation. We then record the two estimates of Contreras Hip–Zhuang used to amplify qualitative information: shell independence and a coarse comparison for the localized approximation.

2.1 Recovery of paths with a buffer

Lemma 2.1. *Let d_n, d be continuous length metrics on $\mathbb{R}^d$, inducing the Euclidean topology, and suppose $d_n \to d$ locally uniformly. Let $F \Subset U$, with F compact and U connected and open. For each $a, \eta > 0$, eventually in n,*

$$d_n(x, y; U) \le (1+\eta) d(x, y; F)\,, \tag{2.1}$$

$$d(x, y; U) \le (1+\eta) d_n(x, y; F)\,, \tag{2.2}$$

for all $x, y \in F$ with $|x-y| \ge a$. Moreover, $\sup_n \operatorname{diam}(F; d_n(\cdot,\cdot;U)) < \infty$ after discarding finitely many n, and for every deterministic $\rho_n \downarrow 0$,

$$\sup_{\substack{x,y\in F\\ |x-y|\le\rho_n}} d_n(x, y; U) \longrightarrow 0\,. \tag{2.3}$$

Proof. Choose a compact neighborhood F' of F contained in U. The d-distance from F to the boundary of F' is positive. Local uniform convergence gives a common positive lower bound for the corresponding d_n-distances. Therefore, sufficiently short almost-minimizing paths for either metric, starting in F, remain in F'.

Let $e_n = \sup_{F'\times F'} |d_n - d|$, and choose $\delta_n \downarrow 0$ with $e_n/\delta_n \to 0$. A d-rectifiable path $P \subset F$ of length L can be subdivided into at most $L/\delta_n + 1$ pieces of length at most δ_n. Replace each piece by a d_n-path with the same endpoints and length at most the original piece length plus $e_n + \delta_n^2$. The common exit barrier places every replacement in U. Their total length is at most

$$L + (L/\delta_n + 1)(e_n + \delta_n^2) = L + o(1)(1+L)\,. \tag{2.4}$$

The same construction, with the two metrics interchanged, applies uniformly to d_n-paths in F. A path with compact image in U has the same length for a global metric and its internal metric on U: sufficiently fine subdivisions use only short distances for which an exit barrier prevents a shorter competitor from leaving U.

The internal metric $d(\cdot,\cdot;U)$ is finite and continuous on $U \times U$. Indeed, sufficiently close points are connected by short global competitors inside U; the set reachable from a fixed point by finite-length paths in U is both open and relatively closed. Connectedness gives finiteness, and the local short-path observation gives continuity. A finite cover of F, together with finitely many paths in U joining the cover centers, gives a uniform bound for the internal diameters for d and, by the same path replacement, for d_n.

Consequently, to prove (2.1)–(2.2) it suffices to consider paths of uniformly bounded length. For endpoint pairs with $|x-y| \ge a$, their global d-distances have a positive minimum; the d_n-distances have half that lower bound for large n. Thus the additive error in (2.4) is absorbed into the factor $1+\eta$. Finally, for sufficiently close endpoints in F, short global almost-minimizers cannot leave U, so the global and internal distances agree. Local uniform convergence and continuity of d prove (2.3). □

The larger open set in this lemma is essential. We do not assert convergence of internal metrics on the same domain, especially when paths can approach its boundary.

2.2 Properness and localization

Lemma 2.2. *Almost surely, D_h is proper: its closed bounded metric balls are compact. In particular, all endpoint pairs admit D_h-geodesics.*

Proof. Let X_r be the infimum of D_h-lengths of paths in $\overline{A}_{r,2r}$ joining its boundary spheres. This variable is strictly positive: such endpoints lie in a compact set and have a positive Euclidean separation. It is finite, since a finite-length path from inside B_r to outside B_{2r} contains such a crossing. By (1.5),

$$r^{-\xi Q} e^{-\xi h_r(0)} X_r \overset{d}{=} X_1 .$$

Choose deterministic radii $r_j \geq e^{j^2}$ increasing so fast that $\mathbb{P}[X_1 < r_j^{-\xi Q/4}] \leq 2^{-j}$. The Gaussian covariance gives $\operatorname{Var} h_r(0) \leq C(1+\log r)$ for $r \geq 1$. Gaussian tails and Borel–Cantelli imply $h_{r_j}(0) \geq -(Q/4)\log r_j$ eventually. A second application of Borel–Cantelli gives

$$X_{r_j} \geq r_j^{\xi Q/2} \longrightarrow \infty \quad \text{eventually, almost surely} .$$

Every path from a fixed Euclidean compact set to $\mathbb{R}^d \setminus B_{2r_j}$ crosses the corresponding annulus for large j. Since D_h is a length metric, its distance from this compact set to infinity tends to infinity. Metric bounded sets are therefore Euclidean bounded. Closed metric balls are Euclidean closed, hence compact. A sequence of almost-minimizing paths stays in a compact metric ball; parameterizing by length and applying Arzelà–Ascoli gives a geodesic. □

Fix a smooth radial $\psi : \mathbb{R}^d \to [0,1]$ equal to one on $B_{1/2}$ and zero outside B_1. Put

$$L_\epsilon = (\log \epsilon^{-1})^{10} , \qquad Z_\epsilon = \int \psi\left(\frac{x}{\epsilon L_\epsilon}\right) K_\epsilon(x)\, dx , \qquad \widehat{h}^*_\epsilon = Z_\epsilon^{-1} h * \left[\psi\left(\frac{\cdot}{\epsilon L_\epsilon}\right) K_\epsilon\right] . \tag{2.5}$$

We only use ϵ small enough that $Z_\epsilon > 0$. Let $\widehat{D}^\epsilon_h$ be defined as in (1.3) with this field, and write

$$N_\epsilon = \mathsf{a}_\epsilon^{-1} D^\epsilon_h , \qquad M_\epsilon = \mathsf{a}_\epsilon^{-1} \widehat{D}^\epsilon_h , \qquad D = D_h . \tag{2.6}$$

The same deterministic normalizer is used for both approximations.

Lemma 2.3. *For every fixed $R < \infty$, the following hold.*

(i) Almost surely, $\sup_{B_R} |\widehat{h}^*_\epsilon - h^*_\epsilon| \to 0$ as $\epsilon \downarrow 0$.

(ii) For every $\eta > 0$, almost surely for all sufficiently small ϵ,

$$\sup_{B_R} |h^*_\epsilon| \leq (\sqrt{2d} + \eta) \log \epsilon^{-1} . \tag{2.7}$$

(iii) For every $a > 0$, almost surely,

$$\sup_{\substack{\epsilon,\delta \in [(n+1)^{-a}, n^{-a}] \\ x \in B_R}} |h^*_\epsilon(x) - h^*_\delta(x)| \longrightarrow 0 . \tag{2.8}$$

The same statements (ii)–(iii) hold with h^*_ϵ replaced by $\widehat{h}^*_\epsilon$.

The proof is given in Appendix A. Part (i) is also [3, Lemma 4.1]. We include the estimates needed to make its rescaled version uniform. The coefficient $\sqrt{2d}$ in (2.7) comes from a Gaussian estimate; integrating the absolute value of a signed kernel would not preserve this coefficient.

Lemma 2.4. *The localized metrics M_ϵ converge to D in probability locally uniformly. Moreover, for each compact F and each $p < 1$, some deterministic $R < \infty$ has the following property for all sufficiently small ϵ: with probability at least p, every pair in F has an M_ϵ-geodesic, and all its minimizing paths are contained in B_R.*

Proof. We use a deterministic observation. Suppose $d_n \to d$ locally uniformly, d is proper, and $\widetilde{d}_n = e^{u_n} \cdot d_n$ with $u_n \to 0$ locally uniformly. Fix a compact F and set $T = 1 + \operatorname{diam}(F; d)$. Choose R with $F \subset B_R$ and $d(F, \partial B_R) > 10T$. For large n, pair distances $d_n(x, y)$ for $x, y \in F$ are less than $2T$, whereas $d_n(F, \partial B_R) > 9T$. Hence almost-minimizers of length less than $2T$ cannot exit B_R. Weighting inside B_R changes lengths by factors between e^{-s_n} and e^{s_n}, where $s_n = \|u_n\|_{\overline{B}_R} \to 0$. An exit for the weighted metric costs at least $9Te^{-s_n}$, while the unweighted competitors have weighted costs at most $2Te^{s_n}$. The strict gap persists. Consequently, for pairs in F, both global metrics can be computed inside B_R and

$$e^{-s_n} d_n(x, y) \le \widetilde{d}_n(x, y) \le e^{s_n} d_n(x, y) \,.$$

This proves $\widetilde{d}_n \to d$ uniformly on $F \times F$ without controlling u_n outside the ball.

Every deterministic sequence of cutoffs has a subsequence on which (1.6) holds almost surely locally uniformly. Lemmas 2.2 and 2.3(i) apply the observation with $d_n = N_\epsilon$ and $u_n = \xi(\widehat{h}^*_\epsilon - h^*_\epsilon)$. The subsequence criterion for convergence in probability proves the first assertion. The same competitor and exit argument, first choosing R on an event of probability greater than p for D and then using convergence in probability on B_R, proves the second assertion. For the smooth positive densities defining M_ϵ, confined almost-minimizers have a minimizing subsequence in a compact ball. □

2.3 Regular variation and the two comparison inputs

Lemma 2.5. *For every fixed $r > 0$,*

$$\frac{\mathsf{a}_{\epsilon/r}}{\mathsf{a}_\epsilon} \longrightarrow r^{\xi Q - 1} \,. \tag{2.9}$$

Consequently $\mathsf{a}_\epsilon = \epsilon^{1 - \xi Q + o(1)}$. For every $a > 0$,

$$\sup_{\epsilon \in [(n+1)^{-a}, n^{-a}]} \left| \frac{\mathsf{a}_\epsilon}{\mathsf{a}_{n^{-a}}} - 1 \right| \longrightarrow 0 \,. \tag{2.10}$$

Proof. A change of variables in (1.3) gives

$$N_\epsilon(ru, rv) = \frac{r\mathsf{a}_{\epsilon/r}}{\mathsf{a}_\epsilon} e^{\xi h_r(0)} N^{h^{0,r}}_{\epsilon/r}(u, v) \,.$$

Use (1.6) for h and for the fixed rescaled field $h^{0,r}$, which has the same law. Equation (1.5), evaluated at $u = 0, v = e_1$, shows that the deterministic ratio $c_\epsilon(r)$ tends to one, since its random limiting point distance is strictly positive.

The lower median a_ϵ is positive, finite, and measurable in ϵ. For measurability, LFPP distances may be expressed using a countable dense family of polygonal paths; their lengths are measurable in the field and cutoff. The lower quantile of the resulting measurable distribution function is measurable. Thus (2.9) says that a is regularly varying at zero, with index $1 - \xi Q$. The representation and uniform convergence theorems for measurable regularly varying functions [1, Theorems 1.2.1 and 1.3.1] give the two conclusions. □

Exact covariance, locality, and Weyl scaling imply that D_h is a weak exponential metric in the sense of [3, Section 1.2], with scale constants $\mathfrak{c}_r = r^{\xi Q}$. For the unperturbed field, normalized rescalings all have the same law. Under a continuous perturbation, bounded weights on each compact set give tight upper moduli; a local exit barrier gives nondegenerate lower moduli. These observations verify the scale condition without assuming it separately. We may therefore use the following two inputs.

Lemma 2.6 (Shell independence)**.** *Fix $q \in (0,1)$ and $q' > 1$. For every $A_0 > 0$ and $b_0 \in (0,1)$, there exists $p < 1$ such that the following holds. Let $r_{i+1}/r_i \le q$, and let E_i be measurable with respect to*

$$\sigma\big((h - h_{r_i}(z))|_{A_{r_i/q',q'r_i}(z)}\big)\,.$$

If $\mathbb{P}[E_i] \ge p$ for every i, then

$$\mathbb{P}\left[\sum_{i=1}^{n} \mathbf{1}_{E_i} < b_0 n\right] \le e^{-A_0 n}\,.$$

This is [3, Lemma 2.7]; translation and constant invariance give the stated center z.

Lemma 2.7 (Localized coarse comparison)**.** *For each $q \in (0,1)$ there are deterministic $C_q \ge 2$ and $\beta_q > 0$, independent of the bounded connected open set U, such that, with probability at least $1 - O_U(\epsilon^{\beta_q})$, simultaneously for all $x, y \in U$,*

$$M_\epsilon(B_{\epsilon^q}(x), B_{\epsilon^q}(y); B_{\epsilon^q}(U)) \le C_q D(x,y;U)\,, \tag{2.11}$$

$$D(B_{\epsilon^q}(x), B_{\epsilon^q}(y); B_{\epsilon^q}(U)) \le C_q M_\epsilon(x,y;U)\,. \tag{2.12}$$

Proof. Use $\zeta = 1 - q$ in [3, Lemma 4.4]. The proof of that lemma, specifically the event in its equation (4.11), gives polynomially high probability, uniformly over the deterministic subset of radii used there. Once that event occurs, both path comparisons are deterministic. Lemma 4.5 of the same paper selects at least half of the radii on which $r\mathsf{a}_{\epsilon/r}/(r^{\xi Q}\mathsf{a}_\epsilon)$ and its reciprocal are bounded by a deterministic constant. Substitution gives (2.11)–(2.12). The covering event in (4.11) covers $B_{\epsilon^{-1}}$, so its exponent and comparison constants are independent of the fixed set U. □

The localization is important in this statement. Proposition 4.2 of [3] states convergence of the probability to one for the original convolution metric; we use the polynomial probability bound for the localized metric from the proof just identified. Lemma 2.3(i) is not assigned a polynomial failure rate.

3 Local comparisons on annuli

A comparison between two fixed annular endpoints follows qualitatively from convergence in probability. We need it for every possible geodesic crossing, together with short connections to nearby endpoints. A buffer around the annulus permits these comparisons to be obtained from local uniform convergence alone.

Fix $\alpha = 3/4$. For $z \in \mathbb{R}^d$ and $r > 0$, write

$$F_{z,r} = \overline{A}_{\alpha r, r}(z)\,, \qquad U_{z,r} = A_{r/4,2r}(z)\,.$$

For a metric d, let $\chi_d(z,r)$ be the infimum of lengths of paths in $F_{z,r}$ from $S_{\alpha r}(z)$ to $S_r(z)$. This quantity is finite and strictly positive for $d = D, M_\epsilon$.

Fix $0 < u < q_0 < 1$, put $v = u/2$, and let

$$\mathcal{N}_\epsilon(u) = \{256^{-k} : k \in \mathbb{Z}\} \cap [\epsilon^u, \epsilon^v]\,, \qquad b_\epsilon = 8\epsilon^{q_0}\,. \tag{3.1}$$

In the definitions below, $r \in \mathcal{N}_\epsilon(u)$ and ϵ is small enough that $b_\epsilon + \epsilon L_\epsilon < r/16$.

Definition 3.1. For $A \geq 1$ and $\eta > 0$, the event $E_\epsilon(z, r)$ requires the following three properties. In these inequalities, $c = c_\epsilon(r)$ and χ_D, χ_M denote $\chi_D(z, r), \chi_{M_\epsilon}(z, r)$.

(i) For every $x \in S_{\alpha r}(z)$ and $y \in S_r(z)$,

$$M_\epsilon(x, y; U_{z,r}) \leq (1 + \eta)cD(x, y; F_{z,r}) \,, \tag{3.2}$$

$$D(x, y; U_{z,r}) \leq (1 + \eta)c^{-1}M_\epsilon(x, y; F_{z,r}) \,. \tag{3.3}$$

(ii) The internal sphere diameters satisfy

$$\operatorname{diam}(S_{\alpha r}(z); D(\cdot, \cdot; U_{z,r})) \leq A\chi_D \,, \qquad \operatorname{diam}(S_{\alpha r}(z); M_\epsilon(\cdot, \cdot; U_{z,r})) \leq A\chi_M \,. \tag{3.4}$$

(iii) For every $x \in F_{z,r}$ and $w \in B_{b_\epsilon}(x)$,

$$M_\epsilon(x, w; U_{z,r}) \leq \eta c\chi_D \,, \qquad D(x, w; U_{z,r}) \leq \eta c^{-1}\chi_M \,. \tag{3.5}$$

Lemma 3.2. *For every $p < 1$ there is $A = A(p, d, \xi, K, D) \geq 1$ such that, for every $\eta > 0$ and $0 < u < q_0 < 1$,*

$$\liminf_{\epsilon \downarrow 0} \inf_{z \in \mathbb{R}^d} \inf_{r \in \mathcal{N}_\epsilon(u)} \mathbb{P}[E_\epsilon(z, r)] \geq p \,. \tag{3.6}$$

The constant A does not depend on η, u, q_0. The event is measurable, up to completion, with respect to

$$\sigma\big((h - h_r(z))|_{A_{r/8,8r}(z)}\big) \,. \tag{3.7}$$

Proof. Put $\delta = \epsilon/r$ and $g = h^{z,r}$. Normalize $D(z + r\cdot, z + r\cdot)$ by $r^{\xi Q}e^{\xi h_r(z)}$. By exact covariance, this gives D_g, with the same law as D. Normalize $M_\epsilon(z + r\cdot, z + r\cdot)$ by $c_\epsilon(r)r^{\xi Q}e^{\xi h_r(z)}$. This gives a_δ^{-1} times the exponential metric of

$$Z_\epsilon^{-1} g * \left[K_\delta \psi \left(\frac{\cdot}{\delta L_\epsilon} \right) \right] \,. \tag{3.8}$$

This is not exactly the cutoff in (2.5) at scale δ: its logarithmic factor is L_ϵ, not L_δ.

The spherical-average bound in [3, Lemma 2.13], radial integration, and the decay of K show that the difference between (3.8) and $g * K_\delta$, uniformly on a fixed ball, is bounded by

$$C(g)(1 + \log \epsilon^{-1})L_\epsilon^{1-d} \,. \tag{3.9}$$

Here $C(g)$ is finite almost surely and has the same law for every z, r; Appendix A gives the calculation. This bound tends to zero in probability uniformly over the indicated z, r. Also $\delta \to 0$ uniformly over the window. To verify the resulting metric convergence uniformly, suppose that a deterministic sequence (ϵ_j, r_j, z_j) violates it. In law, every normalized tuple can be realized using one pinned field g and the two-parameter convolution in (3.8). On this realization, (3.9) tends to zero locally almost surely, and $\mathsf{a}_{\delta_j}^{-1} D_g^{\delta_j} \to D_g$ in probability by (1.6). Apply the proper-limit localization argument of Lemma 2.4 to any almost-surely converging subsequence. This contradicts the proposed violation. Thus the two normalized metrics converge jointly in probability to (D_g, D_g) on each fixed compact set, uniformly over the window. No spatial union bound is applied to (3.9).

Lemma 2.1 applied to $\overline{A}_{\alpha,1} \Subset A_{1/4,2}$ proves that (i) holds with probability tending to one. Normalized annular crossing distances stay bounded away from zero in probability, since they dominate global distances between two compact spheres separated in Euclidean distance. Normalized internal sphere diameters are tight by Lemma 2.1. Hence their ratios to the crossing distances

are tight. Choose A so that (ii) fails with probability less than $(1-p)/3$ in the limit. This choice uses only the limiting pair and not η, u, q_0. Finally,

$$\frac{b_\epsilon}{r} \le 8\epsilon^{q_0-u} \longrightarrow 0\,.$$

The internal modulus in Lemma 2.1, on a slightly larger compact annulus, and the positive crossing lower bounds prove (iii) with probability tending to one. Choosing the diameter error with a strict margin gives (3.6).

Both internal metrics on $U_{z,r}$ are continuous coordinates determined by the field in $A_{r/8,8r}(z)$: for D this is locality, and for M_ϵ it follows from the support radius ϵL_ϵ. Paths in $F_{z,r}$ have the same length in the global and internal coordinates, by the last observation in the proof of Lemma 2.1. To check measurability, rewrite each upper bound involving an infimum over crossings as the corresponding upper bound for the length of every crossing path. The failure of any of (i)–(iii) is then existence of endpoints and one crossing path violating a Borel inequality. Path length is Borel on the product of the continuous-metric and continuous-path spaces, since it is the supremum of sums over rational partitions; evaluation of an internal metric at endpoints is continuous. Thus the failure sets are analytic, and the good event is universally measurable. This argument uses a single existential projection, not an alternating path quantifier. Adding a constant to the field multiplies both metrics and all crossing lengths by the same factor. The event is therefore measurable, after completion, with respect to the field modulo constants in (3.7). □

Lemma 3.3. *There is a choice of $A \ge 1$, depending only on d, ξ, K, D, with the following property. Fix $\eta > 0$, $0 < u < q_0 < 1$, and $R < \infty$. If $\mathcal{R}_\epsilon \subset \mathcal{N}_\epsilon(u)$ is deterministic and $|\mathcal{R}_\epsilon| \ge |\mathcal{N}_\epsilon(u)|/4$, then, with probability at least $1 - O_R(\epsilon^u)$, every point of*

$$\frac{\epsilon^u}{100\sqrt{d}}\mathbb{Z}^d \cap B_{R+1}$$

has a radius $r \in \mathcal{R}_\epsilon$ for which $E_\epsilon(z,r)$ occurs. The estimate is uniform in the deterministic subset $\mathcal{R}_\epsilon$. The choice of A is independent of η, u, q_0, R; the cutoff threshold and implied constant may depend on the fixed parameters η, u, q_0, R.

Proof. Apply Lemma 2.6 with $q = 1/256$, $q' = 8$, $b_0 = 1/2$, and $A_0 = 32(d+2)\log 256$. Choose the marginal probability in Lemma 3.2 strictly larger than the p required there, and fix its A. For small ϵ,

$$|\mathcal{R}_\epsilon| \ge \frac{u \log \epsilon^{-1}}{16 \log 256}\,.$$

Thus at a fixed center the probability that there is no good radius is at most $\epsilon^{2(d+2)u}$. There are $O_R(\epsilon^{-du})$ mesh points. A union bound proves the assertion, with room to spare in the stated exponent. All marginal statements and all estimates are uniform over deterministic subsets of the prescribed size. □

The constant A is now fixed for the rest of the proof. Define

$$\theta = \frac{A}{A+1} \in (0,1)\,. \tag{3.10}$$

The fact that this number does not approach one as the requested comparison accuracy improves is the reason the iteration closes.

4 Improving the comparison constant

The next argument compares the two metrics along a geodesic for one of them. A crossing of a good annulus admits a replacement at the nearly sharp cost in (3.2). The gaps are handled using an earlier coarse comparison. The internal sphere diameter bounds each gap by a fixed multiple of the next crossing, so only a fixed fraction of the total length is charged at the earlier constant.

Fix $R \geq 2$. For $C > 0$ and $q \in (0,1)$, let $\mathcal{C}_{\epsilon,R}^{D\to M}(C,q)$ mean that, for every D-minimizing path P for the internal metric on B_{R+1} whose image lies in B_R, with endpoints x, y,

$$M_\epsilon(B_{8\epsilon^q}(x), B_{8\epsilon^q}(y); B_{R+2}) \leq C \operatorname{len}(P; D)\,. \tag{4.1}$$

Define $\mathcal{C}_{\epsilon,R}^{M\to D}(C,q)$ by interchanging the two metrics. Subsegments of the paths in these properties are again admissible. The fixed larger output domain B_{R+2} is retained at every step of the iteration.

Proposition 4.1. *Fix $\eta > 0$ and $0 < u < q_0 < 1$, put $v = u/2$, and choose a deterministic $\mathcal{R}_\epsilon \subset \mathcal{N}_\epsilon(u)$ of size at least $|\mathcal{N}_\epsilon(u)|/4$. Suppose the covering event of Lemma 3.3 occurs and $\mathcal{C}_{\epsilon,R}^{D\to M}(C,q_0)$ holds. Set $m = \max_{r\in\mathcal{R}_\epsilon} c_\epsilon(r)$. Then, for all sufficiently small ϵ,*

$$\mathcal{C}_{\epsilon,R}^{D\to M}\big(\theta \max\{C,m\} + (1-\theta+3\eta)m,\ v\big) \tag{4.2}$$

holds. The same implication holds with the two metrics interchanged and $m = \max_{r\in\mathcal{R}_\epsilon} c_\epsilon(r)^{-1}$.

Proof. Parameterize an admissible D-minimizing path $P : [0,T] \to B_R$ by D-length, with endpoints x, y. Put $r_{\max} = \epsilon^v$ and $r_{\min} = \epsilon^u$. If $|x-y| \leq 8r_{\max}$, the two target balls intersect and their set distance is zero. Assume otherwise.

Step 1: Separate the endpoint pieces. Let T_y be the first time that P hits $\overline{B}_{2r_{\max}}(y)$. Let t_0 be the last time before T_y at which it hits $\overline{B}_{2r_{\max}}(x)$. For $t \in [t_0, T_y)$, both endpoints are outside any radius-at-most- $r_{\max}$ ball centered at a closest point of the mesh in Lemma 3.3 to $P(t)$.

Suppose $t_{j-1} < T_y$ is defined. Choose such a closest mesh point z_j and a good radius $r_j \in \mathcal{R}_\epsilon$. The mesh error is at most $r_{\min}/100$. Let t_j be the first time after t_{j-1} that P hits $S_{r_j}(z_j)$, and let s_j be its last hit of $S_{\alpha r_j}(z_j)$ before t_j. The outer hit exists because y lies outside this ball. Stop at the first N such that $t_N \geq T_y$. Each step requires a Euclidean displacement at least $r_{\min}/2$, so uniform continuity of P makes N finite.

Write

$$B_j = s_j - t_{j-1}\,, \qquad G_j = t_j - s_j\,, \qquad B = \sum_{j=1}^{N} B_j\,, \qquad G = \sum_{j=1}^{N} G_j\,. \tag{4.3}$$

The path $P|_{[s_j,t_j]}$ is contained in F_{z_j,r_j}, so $G_j \geq \chi_D(z_j, r_j)$.

Step 2: Bound the fraction charged at the old constant. Since x is outside $B_{r_j}(z_j)$, the path has a hit $q_j \leq t_{j-1}$ of $S_{\alpha r_j}(z_j)$. Geodesicity and the sphere-diameter condition give

$$B_j \leq s_j - q_j \leq \operatorname{diam}(S_{\alpha r_j}(z_j); D(\cdot,\cdot; U_{z_j,r_j})) \leq A\chi_D(z_j,r_j) \leq AG_j\,. \tag{4.4}$$

The competitor used here lies in B_{R+1} when ϵ is small, so minimizing in that ambient domain is sufficient. Summing yields

$$B \leq \theta(B+G)\,, \qquad B + G = t_N - t_0 \leq T\,. \tag{4.5}$$

Step 3: Replace crossings and join the gaps. By (3.2), the jth crossing has a replacement in U_{z_j,r_j} of M_ϵ-length at most $(1+\eta)mG_j$, up to an arbitrarily small error. Apply the old

comparison (4.1) to the gap $P|_{[t_{j-1},s_j]}$. It gives a path of length at most CB_j from a b_ϵ-ball around its first endpoint to a b_ϵ-ball around its second endpoint, inside B_{R+2}.

At the inner endpoint of each crossing, use (3.5) to connect to the nearby endpoint of the preceding gap replacement. At the outer endpoint of each crossing except the last, use the same condition to connect to the nearby start of the next gap replacement. There are at most two such connections charged to each crossing; their combined cost is at most $2\eta mG$.

The first gap replacement starts within $b_\epsilon + 2r_{\max}$ of x. Also, $P(T_y)$ and $P(t_N)$ lie in the same closed outer ball, so $|P(t_N) - y| \le 4r_{\max}$. Thus the concatenated path has its endpoints in the target $8r_{\max}$-balls. Its image is in B_{R+2}. Letting the finitely many approximation errors tend to zero gives a total cost bounded by

$$CB + mG + 3\eta mG \le \big[\theta \max\{C, m\} + (1 - \theta + 3\eta)m\big](B + G)\,. \tag{4.6}$$

For the last inequality, replace C by $\max\{C, m\}$ before using (4.5); the coefficient of B is then nonnegative. Equations (4.5)–(4.6) prove the assertion. Interchanging the metrics replaces $c_\epsilon(r)$ by its reciprocal in all three local conditions, so the same construction proves the reverse implication. □

5 Normalization windows and sharp comparison

We now remove the factor $c_\epsilon(r)$ without claiming that it is close to one at every radius. If many separated windows had a substantial fraction of ratios strictly below one, repeated contraction would make M_ϵ strictly smaller than its own probability limit. Ratios strictly above one are ruled out by reversing the comparison. The remaining windows permit an arbitrarily accurate comparison.

Proposition 5.1. *For each $\delta \in (0,1)$ there are $q_\delta \in (0,1)$ and $\beta_\delta > 0$ such that, for every fixed $R \ge 2$, with probability at least $1 - O_R(\epsilon^{\beta_\delta})$, both*

$$\mathcal{C}_{\epsilon,R}^{D\to M}(1+\delta, q_\delta) \quad \text{and} \quad \mathcal{C}_{\epsilon,R}^{M\to D}(1+\delta, q_\delta) \tag{5.1}$$

hold. The two exponents are independent of R.

Proof. Use Lemma 2.7 with $q = 1/2$ and $U = B_{R+1}$, enlarging the endpoint balls and the output domain. With polynomially high probability, both comparison properties hold with $q = 1/2$ and a deterministic $C_0 \ge 2$ independent of R.

Step 1: Choose the contraction parameters. Choose $t \in (0, 1/4)$ and then $\eta > 0$ so small that, with $m_- = 1 - t$ and $m_+ = (1-t)^{-1}$,

$$m_+\left(1 + \frac{3\eta}{1-\theta}\right) < 1 + \frac{\delta}{2}\,, \qquad m_-\left(1 + \frac{3\eta}{1-\theta}\right) < 1 - \frac{t}{2}\,. \tag{5.2}$$

Then choose an integer N so large that

$$\theta^N C_0 < \min\{\delta/2, t/4\}\,. \tag{5.3}$$

Set $v_0 = 1/2$, and for $1 \le i \le 3N$ define $u_i = 3v_{i-1}/4$ and $v_i = u_i/2$. Window i is $\mathcal{N}_\epsilon(u_i)$. They are ordered from smaller to larger spatial radii. Any increasing subsequence of windows is admissible in Proposition 4.1, since its next u_i is less than the previous output exponent.

For a fixed upper bound $m \in \{m_-, m_+\}$ on the ratios used at each step, the constants in Proposition 4.1 can be bounded by the recursion

$$C_{j+1} = \theta C_j + (1 - \theta + 3\eta)m\,, \qquad C_0 \geq 2\,. \tag{5.4}$$

Indeed, $C_0 \geq m$ and this recursion keeps $C_j \geq m$. Its solution is

$$C_N = \theta^N C_0 + m\left(1 + \frac{3\eta}{1-\theta}\right)(1 - \theta^N)\,. \tag{5.5}$$

There are finitely many choices of N windows and finitely many possible pairs of consecutive selected windows, hence finitely many old exponents q_0 to use in Lemma 3.3. For every deterministic selected collection and every allowed deterministic subset of its radii, the union of its covering failures and the initial coarse failure has probability

$$O_R\left(\epsilon^{\beta_{1/2}} + \sum_{i=1}^{3N} \epsilon^{u_i}\right) = O_R(\epsilon^{\min\{\beta_{1/2}, u_{3N}\}})\,. \tag{5.6}$$

The implied constant and the cutoff threshold can be chosen uniformly over the finitely many window choices. The marginal and covering bounds already give uniformity over their deterministic subsets.

Step 2: Exclude too many small ratios. Call a window low if at least a quarter of its radii satisfy $c_\epsilon(r) \leq 1 - t$. We claim that, for all sufficiently small ϵ, fewer than N windows are low. Otherwise take a deterministic sequence $\epsilon_k \downarrow 0$ for which N such windows exist, selecting their low-ratio subsets deterministically. Apply the forward contraction along these windows with $m = m_-$. Equations (5.2)–(5.5) give a constant $\lambda < 1 - t/4$ and, with probability tending to one, the forward comparison property with constant λ and endpoint radius at most $8\epsilon_k^{v_{3N}}$.

Choose R so large that a D-geodesic from 0 to e_1 is contained in B_R with probability greater than 3/4; Lemma 2.2 permits this. On this event and the comparison event,

$$M_{\epsilon_k}(B_{8\epsilon_k^{v_{3N}}}(0), B_{8\epsilon_k^{v_{3N}}}(e_1)) \leq \lambda D(0, e_1)\,. \tag{5.7}$$

By Lemma 2.4, the left side converges in probability to $D(0, e_1) > 0$. Its ratio to $D(0, e_1)$ therefore converges in probability to one. This contradicts the lower bound on the probability of (5.7).

Step 3: Exclude too many large ratios. Call a window high if at least a quarter of its radii satisfy $c_\epsilon(r) \geq (1 - t)^{-1}$. If N high windows existed at arbitrarily small cutoffs, apply the reverse contraction to their subsets, so that $c_\epsilon(r)^{-1} \leq m_-$. Choose R using Lemma 2.4, so that an M_ϵ-geodesic from 0 to e_1 exists and stays in B_R with probability greater than 3/4 for all small ϵ. The resulting inequality is

$$D(B_{8\epsilon_k^{v_{3N}}}(0), B_{8\epsilon_k^{v_{3N}}}(e_1)) \leq \lambda M_{\epsilon_k}(0, e_1)\,, \qquad \lambda < 1\,.$$

The two sides, without λ, both converge in probability to the same positive variable $D(0, e_1)$. This is again a contradiction. Hence fewer than N windows are high for all small cutoffs.

Step 4: Iterate on the remaining windows. Of the $3N$ windows, at least N are neither low nor high. In each of these, more than half of the radii satisfy

$$1 - t < c_\epsilon(r) < (1 - t)^{-1}\,.$$

Select N such windows and all their radii in this interval. These choices are deterministic, since the normalization constants are deterministic. Iterate in both directions with $m = m_+$. By (5.2)–(5.5), the final constant is less than $1 + \delta$. Enlarge the endpoint balls to exponent $q_\delta = v_{3N}$ if the last selected window ends earlier. Equation (5.6) proves (5.1), with $\beta_\delta = \min\{\beta_{1/2}, u_{3N}\} > 0$. □

This proposition is a sharp comparison between small endpoint balls, not an additive pointwise convergence rate. Endpoint control is established next. In particular, no polynomial bound for $\sup|M_\epsilon - D|$ is being inferred from qualitative convergence in probability.

6 Endpoint control and completion of the proof

The radii in Proposition 5.1 can be much larger than the cutoff. A straight Euclidean segment across such a ball need not have small normalized LFPP length. We first connect endpoints at a much smaller radius, where the subcritical inequality gives a power gain. The coarse comparison then transfers the limiting metric's continuity to all shrinking endpoint balls.

Choose $\eta_0 \in (0, Q - \sqrt{2d})$ and $q_* \in (0,1)$ sufficiently close to one that

$$\nu = q_* - 1 + \xi\big(Q - \sqrt{2d} - \eta_0\big) > 0\,. \tag{6.1}$$

Let $C_* = C_{q_*}$ and $\beta_* = \beta_{q_*}$ be from Lemma 2.7.

Lemma 6.1. *Fix $a > 0$ with $a\beta_* > 1$, and put $\epsilon_n = n^{-a}$. Almost surely, for every integer $R \geq 1$, eventually in n,*

$$M_{\epsilon_n}(x, y; B_{R+2}) \leq C_* D(x, y; B_{R+1}) + o_n(1)\,, \qquad x, y \in \overline{B}_R\,, \tag{6.2}$$

where the error is uniform over the endpoints. Consequently, for every sequence $\rho_n \downarrow 0$,

$$\sup_{\substack{x,y\in\overline{B}_R \\ |x-y|\leq\rho_n}} M_{\epsilon_n}(x, y; B_{R+2}) \longrightarrow 0 \quad \textit{almost surely}\,. \tag{6.3}$$

The upper diameters in (6.2) are bounded uniformly in n.

Proof. By Lemmas 2.3 and 2.5, on each fixed ball the M_ϵ-length of any straight segment of Euclidean length at most $2\epsilon^{q_*}$ is at most

$$2\epsilon^{q_*}\mathsf{a}_\epsilon^{-1}\exp\big(\xi \sup_{B_{R+2}} \widehat{h}_\epsilon^*\big) \leq \epsilon^{\nu/2} \tag{6.4}$$

for all sufficiently small ϵ, almost surely. The exponent is positive precisely by (6.1).

Apply (2.11) with $q = q_*$ and $U = B_{R+1}$. Along ϵ_n, its failure probabilities are summable. Borel–Cantelli, and then a countable intersection over R, give the coarse comparison eventually almost surely. Add straight segments from x, y to the nearby endpoints of a path supplied by that comparison. These paths all lie in B_{R+2}, and (6.4) bounds the two extra costs. This proves (6.2). The internal metric $D(\cdot,\cdot; B_{R+1})$ is finite and continuous, hence bounded and uniformly continuous on $\overline{B}_R \times \overline{B}_R$, as shown in Lemma 2.1. The remaining assertions follow. □

Lemma 6.2. *Fix a as in Lemma 6.1. Almost surely, for each integer $k \geq 1$, some finite random R has the following property: for every sufficiently small ϵ, every pair of points in $\overline{B}_k$ has minimizing paths for M_ϵ and N_ϵ, and all such minimizing paths are contained in B_R.*

Proof. First consider $\epsilon_n = n^{-a}$. On the probability-one event in Lemma 6.1, for large n, pairs in $\overline{B}_k$ have competitors inside B_{k+2} of M_{ϵ_n}-length at most

$$T_k = C_* \operatorname{diam}(\overline{B}_k; D(\cdot,\cdot; B_{k+1})) + 1 < \infty\,. \tag{6.5}$$

By properness of D, choose an integer $R > k+4$ so large that

$$D(\overline{B}_{k+1}, \mathbb{R}^d \setminus B_{R-1}) > 4C_*T_k\,. \tag{6.6}$$

The reverse coarse comparison (2.12), with $q = q_*$ and $U = B_{R+1}$, also holds eventually along ϵ_n almost surely, simultaneously for every integer R. Thus its use at this random integer is legitimate. For $x \in \overline{B}_k$ and $y \in S_R$, it gives, when $\epsilon_n^{q_*} < 1$,

$$C_*M_{\epsilon_n}(x,y;B_{R+1}) \geq D(B_{\epsilon_n^{q_*}}(x), B_{\epsilon_n^{q_*}}(y); B_{\epsilon_n^{q_*}}(B_{R+1})) > 4C_*T_k\,.$$

Every path starting in $\overline{B}_k$ which reaches S_R pays more than $4T_k$ up to its first hit. Such a path cannot minimize against the competitors of cost at most T_k. Compactness and the positive continuous density then give minimizing paths contained in B_R.

For $I_n = [(n+1)^{-a}, n^{-a}]$, Lemma 2.3(iii) and (2.10) give

$$\sup_{\substack{\epsilon\in I_n\\ x\in\overline{B}_{R+1}}} \left|\log\frac{\mathsf{a}_{\epsilon_n}}{\mathsf{a}_\epsilon} + \xi\big(\widehat{h}^*_\epsilon(x) - \widehat{h}^*_{\epsilon_n}(x)\big)\right| \longrightarrow 0\,. \tag{6.7}$$

The expression inside the absolute value is the logarithm of the ratio of the two normalized length densities. Replacing its first localized field by h^*_ϵ gives the same conclusion by localization, and hence compares N_ϵ to M_{ϵ_n}. Eventually these ratios lie between $1/2$ and 2, and in fact are strictly closer to one. The competitor bound T_k and exit bound $4T_k$ therefore remain strictly separated. Every sufficiently small cutoff, not just the selected sequence, has all its minimizing paths between pairs in $\overline{B}_k$ inside B_R. □

Proof of Theorem 1.2. Fix $\delta \in (0,1)$, and choose an integer $a \geq 1$ with $a\beta_* > 1$ and $a\beta_\delta > 1$. By Proposition 5.1 and Borel–Cantelli, both sharp comparison properties hold eventually along $\epsilon_n = n^{-a}$, almost surely, for every integer R. Work on this event together with those of Lemmas 6.1 and 6.2.

Fix $k \geq 1$. Choose a finite random integer $R > k+4$ containing all D-geodesics and, eventually, all M_{ϵ_n}-geodesics between pairs in $\overline{B}_k$. For D, every such geodesic lies in a single bounded D-ball, since its length and the distances of its endpoints from 0 are uniformly bounded; properness therefore confines all of them. Lemma 6.2 confines the approximate geodesics. Global minimizing paths are also minimizing for the internal metric on B_{R+1}, so Proposition 5.1 applies. Its endpoint balls have radius $8\epsilon_n^{q_\delta} \to 0$. Lemma 6.1, used on $\overline{B}_{k+1}$, removes these balls from the forward comparison with a uniform error tending to zero. Continuity of D removes them from the reverse comparison. Consequently,

$$M_{\epsilon_n}(x,y) \leq (1+\delta)D(x,y) + o_n(1)\,, \tag{6.8}$$

$$D(x,y) \leq (1+\delta)M_{\epsilon_n}(x,y) + o_n(1)\,, \tag{6.9}$$

uniformly for $x, y \in \overline{B}_k$.

Confinement along the full continuum and Lemmas 2.3 and 2.5 give

$$\sup_{\substack{\epsilon\in[(n+1)^{-a},n^{-a}]\\ x,y\in\overline{B}_k}} |M_\epsilon(x,y) - M_{\epsilon_n}(x,y)| \longrightarrow 0\,. \tag{6.10}$$

Indeed, both minimizing paths remain in a common compact ball, where the normalized length densities differ by a multiplicative factor tending uniformly to one; their distances are uniformly

bounded by the competitors in (6.5). Thus (6.8)–(6.9) and (6.10) imply

$$\limsup_{\epsilon\downarrow 0} \sup_{x,y\in\overline{B}_k} |M_\epsilon(x,y) - D(x,y)| \le \delta \max\{T_k, \operatorname{diam}(\overline{B}_k; D)\}\,. \tag{6.11}$$

The finite factor on the right does not depend on δ.

Repeat this argument for $\delta = 2^{-j}$, $j \ge 1$, using a possibly different sufficiently large integer a for each j. Intersect the resulting countably many probability-one events. Sending $j \to \infty$ in (6.11) gives

$$\sup_{x,y\in\overline{B}_k} |M_\epsilon(x,y) - D(x,y)| \longrightarrow 0\,.$$

Finally, localization and confinement compare the lengths defining M_ϵ and N_ϵ on the same compact ball by factors tending to one. Therefore the same convergence holds for N_ϵ. A countable intersection over k proves (1.7). □

A Field estimates for the cutoff interpolation

We prove Lemma 2.3, including the rescaled localization estimate used in Lemma 3.2. The only sharp coefficient needed is the leading variance of the mollified field. This avoids a positivity assumption on the convolution kernel.

Proof of Lemma 2.3. Step 1: Localization. By [3, Lemma 2.13], for every fixed R, there is an almost surely finite C_R such that

$$|h_s(z)| \le C_R(1 + |\log s|)\,, \qquad z \in \overline{B}_R\,,\ s > 0\,. \tag{A.1}$$

Radial integration represents $h * K_\epsilon$ as the integral of $h_s(z)$ against $|S_1|s^{d-1}K_\epsilon(s)\,ds$. These integrals are absolutely convergent under (A.1); the representation follows first for compactly supported smooth radial kernels and then by Gaussian pairing approximation. The tail decay of K gives

$$|Z_\epsilon - 1| \le CL_\epsilon^{1-d}\,, \tag{A.2}$$

$$\begin{aligned}\sup_{B_R} |Z_\epsilon \widehat{h}^*_\epsilon - h^*_\epsilon| &\le C_R \int_{L_\epsilon/2}^{\infty} u^{-d}(1 + \log\epsilon^{-1} + \log u)\,du \\ &\le C_R(1 + \log\epsilon^{-1})L_\epsilon^{1-d}\,.\end{aligned} \tag{A.3}$$

The same radial representation bounds $\sup_{B_R}|h^*_\epsilon|$ by $C_R(1+\log\epsilon^{-1})$. Combining this with (A.2)–(A.3) proves (i). With ϵ in the kernel replaced by $\delta = \epsilon/r$ but truncation factor retained as L_ϵ, the same calculation proves (3.9), since $\log\delta^{-1} \le \log\epsilon^{-1}$ in the windows used there.

Step 2: Gaussian estimates at unit logarithmic scales. Put $H_t(x) = h^*_{e^{-t}}(x)$. The covariance convention (1.1), the unit mass of K, and its logarithmically integrable tail give

$$\operatorname{Var} H_t(x) = t + O_R(1)\,, \qquad x \in B_R\,,\ t \ge 1\,. \tag{A.4}$$

For completeness, the double convolution of $\log|x-y|^{-1}$ contributes t after scaling and a finite constant depending on K. The terms fixing the unit-sphere average are bounded on B_R after convolution. Absolute integrability follows from the stated decay of K and local integrability of the logarithm.

Let $m(s) = \widehat{K}(se_1)$. From (1.2) and [3, Lemmas 2.10–2.11], m and its radial derivatives decrease faster than any power at infinity; near zero, $m(s) = 1 + O(s^d)$ and $(s\partial_s)^j m(s) = O(s^d)$ for $j \geq 1$. The spectral covariance of a zero-mass pairing is a constant times $\int |\zeta|^{-d} |\widehat{\varphi}(\zeta)|^2 \, d\zeta$. It follows by scaling that the variances of

$$e^{-t}\nabla H_t(x)\,, \quad \partial_t H_t(x)\,, \quad e^{-2t}\nabla^2 H_t(x)\,, \quad e^{-t}\nabla\partial_t H_t(x)\,, \quad \partial_t^2 H_t(x) \tag{A.5}$$

are bounded uniformly in x, t. The relevant radial integrals are finite: spatial derivatives insert positive powers of the radius near zero, logarithmic-scale derivatives annihilate the constant term of m, and all terms decrease rapidly at infinity.

Cover B_R by $O_R(e^{dn})$ cubes of side e^{-n}, with centers z. On each cube, the Gaussian process

$$H_{n+s}(z + e^{-n}u) - H_n(z)\,, \qquad 0 \leq s \leq 1\,,\ u \in [-1,1]^d\,,$$

has uniformly bounded variance and a uniformly Lipschitz canonical metric, by (A.5). Its mean supremum is bounded uniformly, and its supremum has a uniform Gaussian tail. One can see the mean bound by dyadic grids in the compact parameter set: at grid level j, there are $O(2^{(d+1)j})$ increments with standard deviations $O(2^{-j})$, whose expected maximum is $O(2^{-j}\sqrt{j+1})$; the sum is finite. The Gaussian concentration inequality for the supremum [2] then bounds the tail by $C\exp(-cs^2)$ after a fixed centering. The same reasoning applies to the process $\partial_t H_t$ on each cell, using the second derivatives in (A.5).

At the cube centers, (A.4) and a Gaussian union bound imply, for each $\eta > 0$,

$$\sum_{n=1}^{\infty} \mathbb{P}\left[\max_z |H_n(z)| > (\sqrt{2d} + \eta/2)n\right] < \infty\,.$$

The within-cell increments exceed $\eta n/2$ with probability at most $O_R(e^{dn - cn^2})$, also summable. Borel–Cantelli proves (2.7), uniformly over all $t \in [n, n+1]$ for large n. The argument for the derivative process gives, almost surely,

$$\sup_{\substack{t\in[n,n+1]\\ x\in B_R}} |\partial_t H_t(x)| = O_R(n)\,. \tag{A.6}$$

The spectral estimates just used hold for derivatives of every fixed order. Their Gaussian moment bounds and the same compact-parameter grid argument give a jointly differentiable version, with the derivatives in (A.5). We choose this version on the countable compact exhaustion, so the integration in the next step is pathwise.

Step 3: Interpolation. The logarithmic cutoff interval corresponding to $[(n+1)^{-a}, n^{-a}]$ has length $a\log(1 + 1/n) = O(1/n)$ and lies at height $O(\log n)$. Integrating (A.6) over it proves (2.8). Localization transfers (ii)–(iii) to $\widehat{h}^*_\epsilon$. All these assertions are on the original field probability space. □

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